

3.8 Root Test Worksheet

Determine whether each of the following series converge or diverge. You may use any theorem from 3.8 or earlier.

1.
$$\sum_{k=1}^{\infty} \frac{(-2)^k}{k^k}$$

2.
$$\sum_{k=1}^{\infty} \arctan(k^2 + 1)$$

3.
$$\sum_{k=1}^{\infty} \frac{1}{k^3 + 2k + 1}$$

$$4. \quad \sum_{k=1}^{\infty} \left(\frac{k}{k+1} \right)^{k^2}$$

$$5. \quad \sum_{k=1}^{\infty} \frac{1}{5^k + 1}$$

$$6. \quad \sum_{k=1}^{\infty} \left(\frac{1-2k}{k+6} \right)^k$$

$$7. \sum_{k=1}^{\infty} \frac{k^{-1/2}}{2 + \sin^2(k)}$$

$$8. \sum_{k=1}^{\infty} \left(\frac{k^2 - 5}{2k^2 + 1} \right)^k$$